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Circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers

Abstract

Embodiments of the invention include a transistor coupled to a memory element, the memory element being in series with a first bistable resistive element that is configured to switch between a first low resistance state and a first high resistance state. A logic circuit is coupled to the transistor via a series connection to a second bistable resistive element, the second bistable resistive element being configured to switch between a second low resistance state and a second high resistance state.

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Background/Summary

BACKGROUND

(1) The present invention generally relates to fabrication methods and resulting structures for integrated circuits (ICs), and more specifically, to fabrication methods and resulting structures configured and arranged to provide a new circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers.

(2) ICs (also referred to as a chip or a microchip) include electronic circuits on a wafer. The wafer is a semiconductor material, such as, for example, silicon or other materials. An IC is formed of a large number of devices, such as transistors, capacitors, resistors, etc., formed in layers of the IC and interconnected with wiring in the back-end-of-line (BEOL) layers of the wafer. on the wafer. Typical ICs are formed by first fabricating individual semiconductor devices using processes referred to generally as the front-end-of-line (FEOL). A metal-oxide-semiconductor field-effect transistor (MOSFET) is a transistor used for amplifying or switching electronic signals. The MOSFET has a source, a drain, and a metal oxide gate electrode. A conventional FET is a planar device where the entire channel region of the device is formed parallel and slightly below the planar upper surface of the semiconducting substrate. In contrast to a planar FET, there are so-called three-dimensional (3D) devices, such as a FinFET device, which is a three-dimensional structure. One type of device that shows promise for advanced integrated circuit products of the future is generally known as a nanosheet transistor. In general, a nanosheet transistor has a fin-type channel structure that includes a plurality of vertically spaced-apart sheets of semiconductor material. A gate structure for the device is positioned around each of these spaced-apart layers of channel semiconductor material.

SUMMARY

(3) Embodiments of the present invention are directed to providing fabrication methods and resulting structures for a new circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers. A non-limiting method includes providing a transistor coupled to a memory element, the memory element being in series with a first bistable resistive element that is configured to switch between a first low resistance state and a first high resistance state. The method includes providing a logic circuit coupled to the transistor via a series connection to a second bistable resistive element, the second bistable resistive element being configured to switch between a second low resistance state and a second high resistance state.

(4) Other embodiments of the present invention implement features of the above-described devices/structures in methods and/or implement features of the methods in devices/structures.

(5) Additional technical features and benefits are realized through the techniques of the present invention. Embodiments and aspects of the invention are described in detail herein and are considered a part of the claimed subject matter. For a better understanding, refer to the detailed description and to the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The specifics of the exclusive rights described herein are particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other features and advantages of the embodiments of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1A is a schematic of a portion of a new circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers according to one or more embodiments of the invention;

(3) FIG. 1B is a schematic of a portion of the new circuit architecture using same the transistor to operate the circuit in a logic mode according to one or more embodiments of the invention;

(4) FIG. 1C is a schematic of a portion of the new circuit architecture using the same transistor to operate the circuit in memory driver mode according to one or more embodiments of the invention;

(5) FIG. 2A is a schematic of a portion of a new circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers according to one or more embodiments of the invention;

(6) FIG. 2B is a schematic of a portion of the new circuit architecture using same the transistor to operate the circuit in logic mode according to one or more embodiments of the invention;

(7) FIG. 2C is a schematic of a portion of the new circuit architecture using the same transistor to operate the circuit in memory driver mode according to one or more embodiments of the invention;

(8) FIG. 3 depicts a cross-sectional view of a portion of an IC under-fabrication after fabrication operations according to one or more embodiments of the invention;

(9) FIG. 4 depicts a cross-sectional view of a portion of an IC under-fabrication after fabrication operations according to one or more embodiments of the invention;

(10) FIG. 5 depicts a cross-sectional view of a portion of an IC under-fabrication after fabrication operations according to one or more embodiments of the invention;

(11) FIG. 6 depicts a cross-sectional view of a portion of an IC under-fabrication after fabrication operations according to one or more embodiments of the invention;

(12) FIG. 7 depicts a cross-sectional view of a portion of an IC under-fabrication after fabrication operations according to one or more embodiments of the invention;

(13) FIG. 8 depicts a cross-sectional view of a portion of an IC under-fabrication after fabrication operations according to one or more embodiments of the invention;

(14) FIG. 9 depicts a cross-sectional view of a portion of an IC under-fabrication after fabrication operations according to one or more embodiments of the invention;

(15) FIG. 10 depicts a cross-sectional view of a portion of an IC under-fabrication after fabrication operations according to one or more embodiments of the invention;

(16) FIG. 11 depicts a cross-sectional view of a portion of an IC as an example configuration for a front side memory element and backside power delivery network with one contact to each transistor according to one or more embodiments of the invention;

(17) FIG. 12 depicts a cross-sectional view of a portion of an IC as an example configuration for a front side memory element and backside power delivery network with one contact to each transistor according to one or more embodiments of the invention;

(18) FIG. 13 depicts a cross-sectional view of a portion of an IC as another example configuration for a front side memory element and backside power delivery network with one contact to each transistor according to one or more embodiments of the invention;

(19) FIG. 14 depicts a cross-sectional view of a portion of an IC as another example configuration for a front side memory element and backside power delivery network with one contact to each transistor according to one or more embodiments of the invention;

(20) FIG. 15 depicts a cross-sectional view of a portion of an IC as an example configuration for a backside memory element and back-end-of-line layer with one contact to each transistor according to one or more embodiments of the invention;

(21) FIG. 16 depicts a cross-sectional view of a portion of an IC as another example configuration for a backside memory element and back-end-of-line layer with one contact to each transistor according to one or more embodiments of the invention;

(22) FIG. 17 depicts a cross-sectional view of a portion of an IC as an example configuration for a backside memory element and back-end-of-line layer with two contacts to each transistor according to one or more embodiments of the invention;

(23) FIG. 18 depicts a cross-sectional view of a portion of an IC as another example configuration for a backside memory element and back-end-of-line layer with two contacts to each transistor according to one or more embodiments of the invention; and

(24) FIG. 19 is a flowchart of a computer-implemented method of providing a new circuit

architecture using the same transistors with dynamic dual functionality for logic and embedded memory drivers according to one or more embodiments of the invention.

DETAILED DESCRIPTION

(25) For the sake of brevity, conventional techniques related to semiconductor device and integrated circuit (IC) fabrication may or may not be described in detail herein. Moreover, the various tasks and process steps described herein can be incorporated into a more comprehensive procedure or process having additional steps or functionality not described in detail herein. In particular, various steps in the manufacture of semiconductor devices and semiconductor-based ICs are well known and so, in the interest of brevity, many conventional steps will only be mentioned briefly herein or will be omitted entirely without providing the well-known process details.

(26) One or more embodiments of the invention provide fabrication methods and resulting structures for a new circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers. One or more embodiments provide a stacked logic and non-volatile memory circuit using the same field-effect transistors for both the logic functions and non-volatile memory drivers. In one example, the IC may be a motherboard where the logic is routed through the back of the wafer via a backside power delivery network while the embedded memory is routed via a standard back-end-of-line structure. In one or more embodiments, the configuration may be reversed such that embedded memory is routed through the back of the wafer via a backside power delivery network while the logic is routed via a standard back-end-of-line structure. As one or more technical solutions and benefits, the circuit with the same transistors utilized from both logic and embedded memory drivers shrinks the required number of devices on a chip for the same area, increases the number of field-effect transistors/memory cells in a fixed chip area, and/or reduces power consumption by leveraging dual functionality of field-effect transistors for both the logic and memory drivers.

(27) Turning now to a more detailed description of aspects of the present invention, FIG. 1A is a schematic of a portion of a new circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers according to one or more embodiments. The circuit includes a field-effect transistor (FET) connected to a memory element (e.g., non-volatile memory element (NVM)) in series with a first threshold switching selector (TSS) along one branch. The circuit includes the FET connected to a second TSS in series with a logic circuit in a backside power delivery network (BSPDN) along another branch. The first TSS and second TSS each have a high resistance state and a low resistance state. The same FET is utilized to operate the circuit in a logic mode and a memory drive mode.

(28) FIG. 1B is a schematic of a portion of the new circuit architecture using the same transistor to operate the circuit in logic mode according to one or more embodiments. In FIG. 1B, the first threshold switching selector (TSS) is in a high resistance state while the second TSS is in a low resistance state, thereby allowing drain current ($I_{sub.D}$) to flow from a negative voltage source (e.g., $-V_{sub.DD}/2$), through the logic circuit (e.g., one or more logic elements in various combinations including OR gates, AND gates, NOR gates, etc.) through the back side power delivery network (BSPDN) layer (e.g., BSPDN layer **1002** in FIG. **10**), through the second TSS in the low resistance state, and through the FET toward a positive voltage source (e.g., $V_{sub.DD}/2$). The first TSS acts a series resistance with the non-volatile memory element and eliminates the risk of parasitic currents through the NVM element to the FET, by blocking electrical current from flowing. The BSPDN includes one or more logic circuit routings to logic circuits for implementing their logic function by inter-connecting transistors, in which the logic elements (e.g., OR gates, AND gates, NOR gates, etc.) are driven (turned on for logic operations) by the electrical current. It is assumed that gate voltage ($V_{sub.G}$) is applied at the gate to turn on the FET (e.g., a positive gate voltage for an N-type FET (NFET) or a negative gate voltage for a P-type FET (PFET)) as known in the art.

(29) FIG. 1C is a schematic of a portion of the new circuit architecture using the same transistor to

operate the circuit in memory driver mode according to one or more embodiments. In FIG. 1C, the first TSS is in a low resistance state while the second TSS is in a high resistance state, thereby allowing electric current (e.g., $I_{\text{sub.NVM}}$) to flow from a negative voltage source ($-V_{\text{sub.DD}}/2$), through the NVM element, and through the FET toward the positive voltage source ($V_{\text{sub.DD}}/2$). The second TSS switching selector acts as a series resistance with the logic circuit and eliminates the risk of parasitic currents from the logic circuit (element) to the FET, by blocking electrical current from flowing. The NVM element is connected to a memory array circuit (row decoder, column decoder, sense amplifier, analog-to-digital converter, etc.) through which electrical current flows. Particularly, the electrical current ($I_{\text{sub.NVM}}$) is used to drive the NVM memory element for write/read operations by passing through one or more element of the memory circuit (not shown). As noted above, it is assumed that gate voltage ($V_{\text{sub.G}}$) is applied at the gate to turn on the FET (e.g., NFET or PFET) as known in the art.

(30) FIG. 2A is a schematic of a portion of a new circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers according to one or more embodiments. FIG. 2B is a schematic of a portion of the new circuit architecture using same the transistor to operate the circuit in logic mode according to one or more embodiments. FIG. 2C is a schematic of a portion of the new circuit architecture using the same transistor to operate the circuit in memory driver mode according to one or more embodiments. The circuit in FIGS. 2A, 2B, and 2C is analogous to the circuit in FIGS. 1A, 1B, and 1C except no first TSS is connected in series with the NVM element in the circuit. In this case, the NVM element is configured to have resistance levels much higher than the low resistance state of the second TSS in order to block parasitic electrical current from flowing through the NVM element, while allowing electrical current to flow through the logic circuit and the second TSS. Also, the resistance levels of the NVM element are lower than the high resistance state of the second TSS, thereby allowing electrical current to flow through the NVM element in such as case. As seen in FIGS. 2A, 2B, and 2C, the circuit includes the FET connected to a memory element (e.g., non-volatile memory element (NVM)) but no first TSS is present along one branch. The circuit includes the FET connected to the second TSS in series with logic circuit in the BSPDN along another branch. The second TSS has a high resistance state and a low resistance state. Again, the same FET is utilized to operate the circuit in both the logic mode and the memory driver mode.

(31) FIG. 2B is a schematic of a portion of the new circuit architecture using same the transistor to operate the circuit in logic mode according to one or more embodiments. In FIG. 2B, even without the first TSS in a high resistance state, the resistance of the NVM element is high such that it will prevent parasitic current leakage through the NVM. The NVM element is designed with resistance states that are higher than the low resistance state of the second TSS in series with the logic circuit. While the second TSS is in a low resistance state, the high resistance of the NVM element blocks parasitic electrical current, thereby allowing drain current ($I_{\text{sub.D}}$) to flow from a negative voltage source (e.g., $-V_{\text{sub.DD}}/2$), through the logic circuit (e.g., one or more logic elements in various combination include OR gates, AND gates, etc.) via the BSPDN, through the second TSS in the low resistance state, and through the FET toward a positive voltage source (e.g., $V_{\text{sub.DD}}/2$). As noted herein, the NVM element has a resistance level, which is higher than the low resistance state of the second TSS, thereby eliminating the risk of parasitic currents from the NVM element to the FET. As noted above, the BSPDN includes one or more logic circuit routings to logic circuits for implementing their logic function by inter-connecting transistors, in which the logic elements (e.g., OR gates, AND gates, NOR gates, etc.) are driven (turned on for logic operations) by the electrical current. Again, it is assumed that gate voltage ($V_{\text{sub.G}}$) is applied at the gate to turn on the FET (e.g., positive gate voltage for an N-type FET (NFET) or a negative gate voltage for a P-type FET (PFET) as known in the art.

(32) FIG. 2C is a schematic of a portion of the new circuit architecture using the same transistor to operate the circuit in memory driver mode according to one or more embodiments. In FIG. 2C, the

second TSS is in a high resistance state that is greater than the resistance levels of the NVM element, thereby allowing electric current (e.g., $I_{\text{sub.NVM}}$) to flow from a negative voltage source ($-V_{\text{sub.DD}}/2$), through the NVM element, and through the FET toward the positive voltage source ($V_{\text{sub.DD}}/2$). The second TSS acts a series resistance with the logic circuit and eliminates the risk of parasitic currents from the logic circuit (element) to the FET, by blocking parasitic electrical current from flowing. As noted herein, the NVM element is connected to a memory array circuit (row decoder, column decoder, sense amplifier, analog-to-digital converter, etc.) through which electrical current flows. Particularly, the electrical current ($I_{\text{sub.NVM}}$) is used to drive the NVM memory element. As noted above, it is assumed that gate voltage ($V_{\text{sub.G}}$) is applied at the gate to turn on the FET (e.g., NFET or PFET) as known in the art.

(33) The threshold switching selector (TSS) is a bistable resistive system. A bistable resistive system is a system exhibiting a unipolar abrupt reversible and electrically triggered resistance switch between two stable resistance states. The bistable resistive system exhibits electrically-induced abrupt resistivity change such as for insulator-to-metal transition (IMT) materials or the threshold switching selector (TSS). The high-resistivity insulating state (HRS) reduces the current flowing through the system ($I_{\text{sub.OFF}}$) while the abrupt resistance switching to the low-resistivity metallic state (LRS) results in negligible reduction of the on-state drive current ($I_{\text{sub.ON}}$).

Insulator-to-metal transition (IMT) materials (including relaxed, strained, 3D-bulk, thin film, 2D-nanosheet, 1D-nanowire) include materials, for example, such as $\text{VO}_{2.2}$, $\text{NbO}_{2.2}$, $\text{Ca}_{2.2}\text{RuO}_{4.4}$, $\text{LaCoO}_{2.3}$, $\text{Ti}_{2.0}\text{O}_{2.3}$, $\text{Ti}_{3.0}\text{O}_{2.5}$, $\text{SmNiO}_{2.3}$, $\text{NdNiO}_{2.3}$, $\text{V}_{2.0}\text{O}_{2.3}$, $\text{V}_{4.0}\text{O}_{7.7}$, $\text{Fe}_{3.0}\text{O}_{4.4}$, etc., and any oxides of the form $\text{ABO}_{2.3}$ Perovskite. Threshold switching selectors (TSS) may be based on the combination of a thin insulator layer with a metal, for example, such as $\text{Ag}/\text{HfO}_{2.2}$, $\text{Cu}/\text{HfO}_{2.2}$, $\text{Ag}/\text{TiO}_{2.2}$, Cu_{x}S , $\text{Ag}/\text{a-Si}$, $\text{AgTe}/\text{TiN}/\text{TiO}_{2.2}/\text{TiN}$, etc.

(34) Now turning to figures of fabricating a semiconductor structure, fabrication processes are provided for forming an integrated circuit (IC) for the new circuit architecture using transistors with dynamic dual functionality for logic and embedded memory drivers. FIGS. 3-18 illustrate cross-sectional views of an IC 300 according to one or more embodiments. Standard semiconductor fabrication techniques can be utilized to fabricate IC 300 as understood by one of ordinary skill in the art. Any suitable lithography processes including deposition techniques and etching techniques can be utilized herein.

(35) FIG. 3 depicts a cross-sectional view of a portion of the IC 300 starting with a wafer where several fabrication processes have been performed. Front-end-of-line (FEOL) transistors are formed using known fabrication processes. Although one or more embodiments illustrate gate-all-around (GAA) nanosheet FETs for explanation purposes, it should be appreciated that embodiments of the invention are not limited to GAA FETs. Embodiments of the invention apply to any other types of transistor architectures, for example, such as planar FETs, FinFETs, nanowire-FETs, vertical transport FET, etc.

(36) The IC 300 includes transistors formed on a wafer or substrate 302. The wafer or substrate 302 may be formed of (pure) silicon and/or silicon doped with impurities. Other suitable materials can be utilized for the substrates. The transistors include channel layers 308 utilized as the channel regions for the FET device. Material of the channel layers 308 may include silicon. The channel layers 308 are coupled to epitaxial materials 316 which are sources and drains. The epitaxial materials 316 may be P-type epitaxial material, resulting in P-type source and drain regions. The epitaxial materials 316 may be N-type epitaxial material, resulting in N-type source and drain regions. Source/drain contacts 318 are formed on (source/drain) epitaxial materials 316. The source/drain contacts 318 can include trench metals along with silicide. The trench metals may include titanium, tungsten, cobalt, etc.

(37) Inner spacers 312 are formed between the channel layers 308. Example materials of the inner spacers 312 may include SiBCN, SiOCN, SiN, SiOC, SiC, etc. The transistors include gate

materials **320**. Gate caps **314** are formed on the gate materials **320**. The gate caps **314** may be an insulating material, dielectric material, etc. Gate contacts (not shown) are formed on the gate materials **320** for applying gate voltage. The IC **300** includes shallow trench isolation (STI) regions **304** and interlayer dielectric (ILD) material **310**. The STI regions **304** may include one or more dielectric materials such as SiO₂. The STI regions **304** may include a low-k or ultralow-k dielectric material. The ILD material **310** (along with other ILD layers herein) may include low-k or ultralow-k dielectric materials. Gate spacer material **306** is formed, and a horizontal layer of the gate spacer material **306** forms a bottom dielectric isolation layer. Example materials of the gate spacer material **306**, including the bottom dielectric isolation layer, may include SiN, SiBCN, SiOCN, SiOC, etc.

(38) Low-k dielectric materials may generally include dielectric materials having a k value of about 3.9 or less, such as silicon dioxide. The ultralow-k dielectric material generally includes dielectric materials having a k value less than 2.5. Unless otherwise noted, all k values mentioned in the present application are measured relative to a vacuum. Exemplary ultralow k dielectric materials generally include porous materials such as porous organic silicate glasses, porous polyamide nanofoams, silica xerogels, porous hydrogen silsesquioxane (HSQ), porous methylsilsesquioxane (MSQ), porous inorganic materials, porous CVD materials, porous organic materials, or combinations thereof. The ultralow-k dielectric material can be produced using a templated process or a sol-gel process as is generally known in the art. In the templated process, the precursor typically contains a composite of thermally labile and stable materials. After film deposition, the thermally labile materials can be removed by thermal heating, leaving pores in the dielectric film. In the sol gel process, the porous low k dielectric films can be formed by hydrolysis and polycondensation of an alkoxide(s) such as tetraethoxysilane (TEOS).

(39) FIG. 4 depicts a cross-sectional view of a portion of the IC **300** under-fabrication after fabrication operations. FIG. 4 illustrates middle-of-line (MOL) processing and/or back-end-of-line (BEOL) processing on the front side of the wafer. In the M1 level (e.g., metal layer 1 level), interconnect structures **420** are formed in ILD material **410**. The interconnect structures **420** are formed in contact with respective source/drain contacts **318**. Each of the interconnect structures **420** includes a metal via **402** connected to a metal line **404**.

(40) FIG. 5 depicts a cross-sectional view of a portion of the IC **300** under-fabrication after fabrication operations. FIG. 5 illustrates forming embedded memory elements and threshold switching selector (TSS) in series. During further MOL processing, the M2 level and M3 level are formed. In the M2 level, a structure **520** is formed in ILD material **510**. The structure **520** includes an NVM bottom contact **522** in contact with the metal line **404** of the interconnect structure **420** and a memory element **524**. The memory elements **524** are two-terminal memory devices. Examples of the memory elements **524** may include magnetoresistive random-access memory (MRAM), spin-transfer torque magnetic random-access memory (STT-MRAM), resistive random access memory (RRAM), phase-change random access memory (PCRAM), conductive bridging random access memory (CBRAM), ferroelectric random access memory (FeRAM), etc.

(41) In the M3 level, a structure **530** is formed on the structure **520**. The structure **530** includes a bistable resistive element such as a TSS device **534**, where the TSS device **534** is sandwiched between metal via **532** and metal via **536**. The structure **530** also includes (M3) metal line **538**. It should be noted that the metal vias **532** and **536** (as well as metal vias **532A** and **536A** and metal vias **1532** and **1536** discussed below) are not simply vias per se. The TSS is integrated in the via itself. Depending on the material used for the bistable resistive element, different types of conductive materials (metals for example) are chosen for the vias to work in conjunction with the switching material. As such, the metal vias **532** and **536** can be different conductive materials.

(42) The vias **402**, (M1) metal lines **404**, an NVM bottom contacts **522**, vias **532**, vias **536**, and (M3) metal lines **538** are formed of electrically conductive materials including metals, metal alloys, etc., as known in the art for MOL and BEOL processing.

(43) The example illustrated in FIG. 5 depicts a 1T-1R configuration which may use MRAM integrated at the M2 level. Although the 1T-1R is illustrated, more than 1 transistor (1T) can be utilized to drive 1 memory element (1R). For example, a 2T-1R configuration can be utilized where 2 transistors are utilized to drive 1 memory element (1R). Other configurations are contemplated. Although the memory elements **524** are shown at the M2 level, the memory elements **524** can be integrated at different MOL/BEOL levels. Similarly, although the TSS devices **534** are illustrated at the M3 level, the TSS devices **534** can be integrated at different MOL/BEOL levels. In one or more embodiments, the memory elements **524** and the TSS devices **534** can be integrated in the same metal level. In one or more embodiments, the memory elements **524** and TSS devices **534** can be implemented on the backside of the wafer as discussed herein.

(44) FIG. 6 depicts a cross-sectional view of a portion of the IC **300** under-fabrication after fabrication operations. As part of the BEOL processing and/or far BEOL processing, FIG. 6 illustrates formation of a BEOL layer **602** that represents routing to memory array circuits (row decoder, column decoder, sense amplifier, analog-to-digital converter, etc.) as understood by one of ordinary skill in the art. The details of the BEOL layer **602** with its memory array circuits are not illustrated for conciseness, but it is understood that the memory array circuits are present and operatively connected in the BEOL layer **602**. For example, at least one or more memory circuits are operatively connected to each of the memory elements **524**, respectively.

(45) For the operation of an individual memory element **524**, FIG. 6 illustrates that the M3 metal line corresponds to the bit line (BL), the M1 metal line **404** corresponds to the source line (SL), and the gate material **320** corresponds to the word line (WL). Accordingly, voltage (or signals) is appropriately applied to the bit line, the source line, and the word line to read and write to the memory element **524** as desired.

(46) FIG. 7 depicts a cross-sectional view of a portion of the IC **300** under-fabrication after fabrication operations. In preparation for future backside power delivery network integration, before flipping the wafer, a sacrificial carrier wafer **702** is attached to the BEOL layer **602**. The carrier wafer **702** may be silicon and/or any other suitable material. The wafer is flipped, and chemical mechanical polishing/planarization (CMP) is performed on the substrate **302** to make the substrate **302** coplanar with the STI regions **304**.

(47) FIG. 8 depicts a cross-sectional view of a portion of the IC **300** under-fabrication after fabrication operations. FIG. 8 illustrates substrate etch back of the substrate **302**, ILD material **802** deposition, and CMP.

(48) FIG. 9 depicts a cross-sectional view of a portion of the IC **300** under-fabrication after fabrication operations. FIG. 9 illustrates backside MOL formation in order to make contact with the S/D epitaxial material **316**. In the backside MOL formation, backside interconnect structures **930** are formed through the bottom dielectric isolation layer to contact the respective S/D epitaxial material **316**. Each backside interconnect structure **930** includes a via **932**, a TSS device **934**, and a via **936**. It should be noted that the metal vias **932** and **936** (as well as metal vias **932A** and **936A** discussed below) are not simply vias per se. The TSS is integrated in the via itself. Depending on the material used for the bistable resistive element, different types of conductive materials (metals for example) are chosen for the vias to work in conjunction with the switching material. As such, the metal vias **532** and **536** can be different conductive materials. Each backside interconnect structure **930** is contacting the same S/D epitaxial material **316** (e.g., the same source/drain region) as a corresponding memory element **524** but on an opposite side of the S/D epitaxial material **316**. In one or more embodiments, the TSS device **934** and the TSS device **534** are formed on opposite sides of the same S/D epitaxial material **316**. In one or more embodiments discussed herein, the TSS device **534** may be omitted according to the examples discussed in FIGS. 2A, 2B, and 2C.

(49) FIG. 10 depicts a cross-sectional view of a portion of the IC **300** under-fabrication after fabrication operations. FIG. 10 illustrates formation of a backside power delivery network (BSPDN) layer **1002**. The BSPDN layer **1002** which includes routing capability to enable logic

circuits using interconnections between transistors as well as power lines. Examples of logic circuits include AND gates, OR gates, NOR gates, etc. The details of the BSPDN layer **1002** are not illustrated so as not to obscure the figure.

(50) FIGS. **11** and **12** depict cross-sectional views of a portion of the IC **300** after flipping the wafer such that the front side is up and removal of the carrier wafer **702**. FIGS. **11** and **12** depict a final configuration for a front side memory element and backside power delivery network with one contact to each FET. Each given transistor can serve two possible functionalities: logic function for a logic circuit and driver for embedded memories. The dynamic functionality switch is based on real-time circuit requirements/needs.

(51) FIG. **11** illustrates the transistor operating in logic function for a logic circuit. In FIG. **11**, power is supplied to a logic circuit of a logic device in the BSPDN layer **1002**. The TSS device **934A** of a backside interconnect structure **930** is in a low resistance state (LRS) and the TSS device **534A** of an interconnect structure **530** is in a high resistance state (HRS), by applying voltage as discussed in FIGS. **1A**, **1B**, and **1C**. As such, electrical current flows from the logic circuit in the BSPDN layer **1002**, through a via **936A**, through the TSS device **934A**, through the via **932A**, into one source/drain epitaxial material **316** of the transistor, through the channel layers **308**, into the other source/drain epitaxial material **316**, out through the source/drain contact **318**, into a via **402A**, and out through the M1 metal line **404A**. Because of the high resistance state of the TSS device **534A** in series with the memory element **524A**, no parasitic electrical current flows through memory element **524A**.

(52) FIG. **12** illustrates the transistor operating as a memory driver. In FIG. **12**, power is supplied to the memory array circuit in the BEOL layer **602**. Unlike FIG. **11**, the TSS device **934A** of a backside interconnect structure **930** is in a high resistance state (HRS) and the TSS device **534A** of an interconnect structure **530** is in a low resistance state (LRS), by applying voltage as discussed in FIGS. **1A**, **1B**, and **1C**. As such, electrical current flows from the memory array circuit in the BEOL layer **602**, through an M3 metal line **538A**, through via **536A**, through the TSS device **534A**, through the via **532A**, into the memory element **524A**, through an NVM bottom contact **522**, through M1 metal line **404B**, through via **402B**, into one source/drain contact **318**, into one source/drain epitaxial material **316** of the transistor, through the channel layers **308**, into the other source/drain epitaxial material **316**, out through another source/drain contact **318**, into a via **402A**, and out through the M1 metal line **404A**. Because of the high resistance state of the TSS device **934A** in series with the logic circuit (of a logic device), no parasitic electrical current flows through logic device in series with the TSS device **934A**.

(53) FIGS. **13** and **14** depict cross-sectional views of a portion of the IC **300** after forming backside interconnect structures **1302** in contact with the source/drain epitaxial material **316** and then flipping the wafer such that the front side is up. The backside interconnect structures **1302** are formed of conductive materials including metals, metal alloys, etc. FIGS. **13** and **14** depict another final configuration for a front side memory element and backside power delivery network with one contact to each FET. Again, the same transistor can serve two possible functionalities: logic function for a logic circuit and driver for embedded memories. The dynamic function switch is based on real-time circuit requirements.

(54) FIG. **13** illustrates the transistor operating in logic function for a logic circuit. In FIG. **13**, power is supplied to a logic circuit of a logic device in the BSPDN layer **1002**. The TSS device **934A** of a backside interconnect structure **930** is in a low resistance state (LRS) and the TSS device **534A** of an interconnect structure **530** is in a high resistance state (HRS), by applying voltage as discussed in FIGS. **1A**, **1B**, and **1C**. As such, electrical current flows from the logic device in the BSPDN layer **1002**, through a via **936A**, through the TSS device **934A**, through the via **932A**, into one source/drain epitaxial material **316** of the transistor, through the channel layers **308**, into the other source/drain epitaxial material **316**, out through the backside interconnect structure **1302**, and back into the BSPDN layer **1002**. Because of the high resistance state of the TSS device **534A** in

series with the memory element **524A**, no parasitic electrical current flows through memory element **524A**.

(55) FIG. **14** illustrates the transistor operating as a memory driver for the memory element **524**. In FIG. **14**, power is supplied to the memory array circuits in the BEOL layer **602**. Unlike FIG. **13**, the TSS device **934A** of a backside interconnect structure **930** is in a high resistance state (HRS) and the TSS device **534A** of an interconnect structure **530** is in a low resistance state (LRS), by applying voltage as discussed in FIGS. **1A**, **1B**, and **1C**. As such, electrical current flows from the memory array circuits in the BEOL layer **602**, through an M3 metal line **538A**, through via **536A**, through the TSS device **534A**, through the via **532A**, into the memory element **524A**, through an NVM bottom contact **522**, through M1 metal line **404B**, through via **402B**, into one source/drain contact **318**, into one source/drain epitaxial material **316** of the transistor, through the channel layers **308**, into the other source/drain epitaxial material **316**, out through another source/drain contact **318**, into a via **402A**, and out through the M1 metal line **404A**. Because of the high resistance state of the TSS device **934A** in series with the logic circuit (of a logic device), no parasitic electrical current flows through logic device in series with the TSS device **934A**.

(56) FIGS. **15** and **16** depict cross-sectional views of a portion of the IC **300**. FIGS. **15** and **16** depict a final configuration for a backside memory element and back-end-of-line with one contact to each FET. Each given transistor can serve two possible functionalities: logic function of a logic circuit and driver for embedded memories. The dynamic function switch is based on real-time circuit requirements. Unlike FIGS. **11** and **12**, the arrangement of the memory elements and logic devices are formed on opposite sides of the wafer in FIGS. **15** and **16**. Particularly, the memory elements **524** are formed on the backside of the wafer while the logic circuits of logic devices are formed on the front side of the wafer.

(57) On the front side of the wafer, interconnect structures **1530** are analogous to the interconnect structures **930**. Each interconnect structure **1530** includes a via **1532**, a TSS device **1534**, and a via **1536**. Each interconnect structure **1530** is connected to a BEOL layer **1580** having the routing of logic circuits for logic devices on the front side of the wafer, analogous to the BSPDN layer **1002**. An interconnect structure **1530** is in series with a logic device in the BEOL layer **1580**.

(58) On the backside of the wafer, interconnect structure **1502** is in contact with the source/drain epitaxial material **316**. A metal line **1504** is in contact with the interconnect structure **1502**. In this configuration, the structure **520** is on the backside, where the an NVM bottom contact **522** is the metal line **1504** and the memory element **524**. For an interconnect structure **1540**, the via **532** contacts the memory element **524**, the TSS device **534** contacts the via **532**, and the via **536** contacts the TSS device **534**. The via **536** is in contact with a backside routing layer **1590** with its memory array circuits. The backside routing layer **1590** is analogous to the BEOL layer **602** with memory array circuits. The backside routing layer **1590** includes a power delivery network.

(59) FIG. **15** illustrates the transistor operating in logic function for a logic circuit. In FIG. **15**, power is supplied to a logic circuit via the BEOL layer **1580**. The TSS device **1534** of a (front side) interconnect structure **1530** is in a low resistance state (LRS) and the TSS device **534** of a (backside) interconnect structure **1540** is in a high resistance state (HRS), by applying voltage as discussed in FIGS. **1A**, **1B**, and **1C**. As such, electrical current flows from the logic device in the routing layer **1580**, through a via **1536**, through the TSS device **1534**, through the via **1532**, into metal line **404**, through via **402**, into one source/drain contact **318**, into one source/drain epitaxial material **316** of the transistor, through the channel layers **308**, into the other source/drain epitaxial material **316**, out through the interconnect structure **1502**, and into the metal line **1504**. Because of the high resistance state of the TSS device **534** in series with the memory element **524**, no parasitic electrical current flows through memory element **524**.

(60) FIG. **16** illustrates the transistor operating as a memory driver for the memory element **524**. In FIG. **16**, power is supplied to the memory array circuits in the routing layer **1590**. Unlike FIG. **15**, the TSS device **1534** of the interconnect structure **1530** is in a high resistance state (HRS) and the

TSS device **534** of an interconnect structure **1540** is in a low resistance state (LRS), by applying voltage as discussed in FIGS. **1A**, **1B**, and **1C**. As such, electrical current flows from the memory array circuits in the routing layer **1590**, through the via **536**, through the TSS device **534**, through the via **532**, into the memory element **524**, through an NVM bottom contact **522**, through metal line **1504**, through interconnect structure **1502**, into one source/drain epitaxial material **316** of the transistor, through the channel layers **308**, into the other source/drain epitaxial material **316**, out through another interconnect structure **1502**, and out through another metal line **1504**. Because of the high resistance state of the TSS device **1534** in series with the logic circuit (of a logic device), no parasitic electrical current flows through logic device in series with the TSS device **1534**.

(61) FIGS. **17** and **18** depict cross-sectional views of a portion of the IC **300**. FIGS. **17** and **18** depict another final configuration for a backside memory element and back-end-of-line with two contacts to each FET. Each given transistor can serve two possible functionalities: logic function of a logic circuit and driver for embedded memories. The dynamic function switch is based on real-time circuit requirements. Unlike FIGS. **13** and **14**, the arrangement of the memory elements and logic devices are formed on opposite sides of the wafer in FIGS. **17** and **18**. Particularly, the memory elements **524** are formed on the backside of the wafer while the logic circuits of logic devices are formed on the front side of the wafer.

(62) On the front side of the wafer (as discussed in FIGS. **15** and **16**), interconnect structures **1530** are analogous to the interconnect structures **930**. Each interconnect structure **1530** includes a via **1532**, a TSS device **1534**, and a via **1536**. Each interconnect structure **1530** is connected to a BEOL layer **1580** having the routing of logic circuits for logic devices on the front side of the wafer, analogous to the BSPDN layer **1002**. An interconnect structure **1530** is in series with a logic device in the BEOL layer **1580**. In addition to the structures in FIGS. **15** and **16**, FIGS. **17** and **18** further include, on the front side of the wafer, an interconnect structure **1720** is in contact with the BEOL layer **1580** and one source/drain contact **318** of the transistor.

(63) On the backside of the wafer (as discussed in FIGS. **15** and **16**), interconnect structure **1502** is in contact with the source/drain epitaxial material **316**. A metal line **1504** is in contact with the interconnect structure **1502**. In this configuration, the structure **520** is on the backside, where the an NVM bottom contact **522** is the metal line **1504** and the memory element **524**. For an interconnect structure **1540**, the via **532** contacts the memory element **524**, the TSS device **534** contacts the via **532**, and the via **536** contacts the TSS device **534**. The via **536** is in contact with a backside routing layer **1590** with its memory array circuits. The backside routing layer **1590** is analogous to the BEOL layer **602** with memory array circuits. The backside routing layer **1590** includes a power delivery network.

(64) FIG. **17** illustrates the transistor operating in logic function for a logic circuit. In FIG. **17**, power is supplied to a logic circuit of a logic device in the BEOL layer **1580**. The TSS device **1534** of a (front side) interconnect structure **1530** is in a low resistance state (LRS) and the TSS device **534** of a (backside) interconnect structure **1540** is in a high resistance state (HRS), by applying voltage as discussed in FIGS. **1A**, **1B**, and **1C**. As such, electrical current flows from the logic device in the routing layer **1580**, through a via **1536**, through the TSS device **1534**, through the via **1532**, into the metal line **404**, through the via **402**, into one source/drain contact **318**, into one source/drain epitaxial material **316** of the transistor, through the channel layers **308**, into the other source/drain epitaxial material **316** of the transistor, out through the via **1722**, through the metal line **1724**, through the via **1726**, and back into the routing layer **1580**. Because of the high resistance state of the TSS device **534** in series with the memory element **524**, no parasitic electrical current flows through memory element **524**.

(65) FIG. **18** illustrates the transistor operating as a memory driver for the memory element **524**. In FIG. **18**, power is supplied to the memory array circuits in the routing layer **1590**. Unlike FIG. **17**, the TSS device **1534** of the interconnect structure **1530** is in a high resistance state (HRS) and the TSS device **534** of an interconnect structure **1540** is in a low resistance state (LRS), by applying

voltage as discussed in FIGS. 1A, 1B, and 1C. As such, electrical current flows from the memory array circuits in the routing layer **1590**, through the via **536**, through the TSS device **534**, through the via **532**, into the memory element **524**, through an NVM bottom contact **522**, through metal line **1504**, through interconnect structure **1502**, into one source/drain epitaxial material **316** of the transistor, through the channel layers **308**, into the other source/drain epitaxial material **316**, out through another interconnect structure **1502**, and out through another metal line **1504**. Because of the high resistance state of the TSS device **1534** in series with the logic circuit (of a logic device), no parasitic electrical current flows through logic device in series with the TSS device **1534**.

(66) It should be appreciated that embodiments are not meant to be limited by use of the front side or backside of the wafer. Rather, illustrations show that the same transistor can be utilized for logic circuits and for driving the memory elements in any configuration, whether the front side or backside, according to one or more embodiments of the invention.

(67) FIG. **19** is a flowchart of a computer-implemented method **1900** of providing a new circuit architecture using the same transistor with dynamic dual functionality for logic and embedded memory drivers according to one or more embodiments. At block **1902**, the method **1900** includes providing a transistor coupled to a memory element **524**, the memory element **524** being in series with a first bistable resistive element (e.g., TSS device **534**) that is configured to switch between a first low resistance state and a first high resistance state. At block **1904**, the method **1900** includes providing a logic circuit (e.g., via layers **1002**, **1580**) coupled to the transistor via a series connection to a second bistable resistive element, the second bistable resistive element (e.g., TSS device **934**, **1534**) being configured to switch between a second low resistance state and a second high resistance state.

(68) The memory element **524** is configured to form a circuit with the transistor in which the circuit excludes/prevents electrical current flow to (although there is a physical connection) and/or electrically isolates the logic circuit (e.g., via layer **1002**, **1580**). The circuit defines the flow of electrical current. The logic circuit (e.g., via layer **1002**, **1580**) is configured to form a circuit with the transistor in which the circuit excludes/prevents electrical current flow to and/or electrically isolates the memory element **524**.

(69) The memory element **524** is coupled to a memory array circuit in a back-end-of-line network (e.g., BEOL layer **602**) on a front side, the front side being above the transistor. The logic circuit is part of a network (e.g., via layer **1002**, **1580**) for routing logic circuits and delivering power. The memory element **524** and the logic circuit are formed on opposite sides of the transistor.

(70) The front side of the wafer (above the transistor) includes a middle-of-the-line (MOL) and back-end-of-the-line (BEOL) electrical network. The back side of the wafer (below the transistor) includes the backside electrical network. In any implementation whether memory on top or bottom, these electrical networks support the memory elements they are routed to enable memory (memory array) operation, and these electrical networks support the transistor they are routed to enable standard transistor operation (i.e., logic function of the logic elements, circuits, and/or devices). Both of them can have power delivery capability.

(71) A circuit with the transistor and the memory element **524** excludes/prevents electrical current flow to and/or electrically isolates any terminal of the transistor that connects to the logic circuit. A circuit with the transistor and the logic circuit excludes/prevents electrical current flow to and/or electrically isolates any terminal of the transistor that connects to the memory element.

(72) The first and second bistable resistive elements (e.g., TSS device **934**, **1534**) each include a unipolar, abrupt, reversible, and electrically triggered resistance switch that operates between two stable resistance states. At least one of the first and second bistable resistive elements (e.g., TSS device **934**, **1534**) includes an insulator-to-metal transition material. The insulator-to-metal transition material includes VO.sub.2, NbO.sub.2, Ca.sub.2RuO.sub.4, LaCoO.sub.3, Ti.sub.2O.sub.3, Ti.sub.3O.sub.5, SmNiO.sub.3, NdNiO.sub.3, V.sub.2O.sub.3, V.sub.4O.sub.7, Fe.sub.3O.sub.4, an oxide of the form ABO.sub.3-perovskite, and/or any combinations thereof.

(73) At least one of the first and second bistable resistive elements (e.g., TSS device **934**, **1534**) includes a threshold-switching selector. The threshold-switching selector includes a combination of a thin insulator layer with a metal such as Ag-doped HfO₂, Cu/HfO₂, Ag/TiO₂, Cu_xS, Ag/a-Si, AgTe/TiN/TiO₂/TiN, W/Cu_xS, and/or any combinations thereof. The memory element includes non-volatile memory (NVM). The memory element **524** includes a charge trap memory, magnetic random access memory (MRAM) (such as, e.g., spin-transfer-torque (STT) MRAM, or spin-orbit torque (SOT) MRAM), ferroelectric-RAM (FRAM), phase-change memory (PCM), resistive-RAM (ReRAM) (such as oxygen vacancy RAM (OxRAM) or conductive-bridge RAM (CBRAM)), and/or any combinations thereof. The transistor is one of a planar field-effect transistor (FET), a finFET, gate-all-around (GAA) FET (e.g., GAA nanowire-FET or GAA nanosheet-FET), a vertical-transport-FET (VTFET), and/or a stacked-FET (where PFET is stacked on NFET or vice versa). Additionally, any of these transistor architectures can be integrated on bulk substrate or silicon-on-insulator (SOI) substrate.

(74) A first mode, to selectively operate the memory element **524**, includes the first bistable resistive element in the first low resistance state and the second bistable resistive element in the second high resistance state. A second mode, to selectively operate the logic circuit, includes the first bistable resistive element in the first high resistance state and the second bistable resistive element in the second low resistance state, the first and second low resistance states permitting a flow of electrical current, the first and second high resistance states limiting a flow of electrical current.

(75) Various embodiments of the present invention are described herein with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of this invention. Although various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings, persons skilled in the art will recognize that many of the positional relationships described herein are orientation-independent when the described functionality is maintained even though the orientation is changed. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the present invention is not intended to be limiting in this respect.

Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship. As an example of an indirect positional relationship, references in the present description to forming layer “A” over layer “B” include situations in which one or more intermediate layers (e.g., layer “C”) is between layer “A” and layer “B” as long as the relevant characteristics and functionalities of layer “A” and layer “B” are not substantially changed by the intermediate layer(s).

(76) The phrase “selective to,” such as, for example, “a first element selective to a second element,” means that the first element can be etched and the second element can act as an etch stop.

(77) As used herein, “p-type” refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing substrate, examples of p-type dopants, i.e., impurities, include but are not limited to: boron, aluminum, gallium and indium.

(78) As used herein, “n-type” refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon containing substrate examples of n-type dopants, i.e., impurities, include but are not limited to antimony, arsenic and phosphorous.

(79) As previously noted herein, for the sake of brevity, conventional techniques related to semiconductor device and integrated circuit (IC) fabrication may or may not be described in detail herein. By way of background, however, a more general description of the semiconductor device fabrication processes that can be utilized in implementing one or more embodiments of the present invention will now be provided. Although specific fabrication operations used in implementing one or more embodiments of the present invention can be individually known, the described combination of operations and/or resulting structures of the present invention are unique. Thus, the unique combination of the operations described in connection with the fabrication of a

semiconductor device according to the present invention utilize a variety of individually known physical and chemical processes performed on a semiconductor (e.g., silicon) substrate, some of which are described in the immediately following paragraphs.

(80) In general, the various processes used to form a micro-chip that will be packaged into an IC fall into four general categories, namely, film deposition, removal/etching, semiconductor doping and patterning/lithography. Deposition is any process that grows, coats, or otherwise transfers a material onto the wafer. Available technologies include physical vapor deposition (PVD), chemical vapor deposition (CVD), electrochemical deposition (ECD), molecular beam epitaxy (MBE) and more recently, atomic layer deposition (ALD) among others. Removal/etching is any process that removes material from the wafer. Examples include etch processes (either wet or dry), and chemical-mechanical planarization (CMP), and the like. Semiconductor doping is the modification of electrical properties by doping, for example, transistor sources and drains, generally by diffusion and/or by ion implantation. These doping processes are followed by furnace annealing or by rapid thermal annealing (RTA). Annealing serves to activate the implanted dopants. Films of both conductors (e.g., poly-silicon, aluminum, copper, etc.) and insulators (e.g., various forms of silicon dioxide, silicon nitride, etc.) are used to connect and isolate transistors and their components. Selective doping of various regions of the semiconductor substrate allows the conductivity of the substrate to be changed with the application of voltage. By creating structures of these various components, millions of transistors can be built and wired together to form the complex circuitry of a modern microelectronic device.

(81) As noted above, atomic layer etching processes can be used in the present invention for via residue removal, such as can be caused by via misalignment. The atomic layer etch process provide precise etching of metals using a plasma-based approach or an electrochemical approach. The atomic layer etching processes are generally defined by two well-defined, sequential, self-limiting reaction steps that can be independently controlled. The process generally includes passivation followed selective removal of the passivation layer and can be used to remove thin metal layers on the order of nanometers. An exemplary plasma-based approach generally includes a two-step process that generally includes exposing a metal such a copper to chlorine and hydrogen plasmas at low temperature (below 20° C.). This process generates a volatile etch product that minimizes surface contamination. In another example, cyclic exposure to an oxidant and hexafluoroacetylacetone (Hhfac) at an elevated temperature such as at 275° C. can be used to selectively etch a metal such as copper. An exemplary electrochemical approach also can include two steps. A first step includes surface-limited sulfidization of the metal such as copper to form a metal sulfide, e.g., Cu.sub.2S, followed by selective wet etching of the metal sulfide, e.g., etching of Cu.sub.2S in HCl. Atomic layer etching is relatively recent technology and optimization for a specific metal is well within the skill of those in the art. The reactions at the surface provide high selectivity and minimal or no attack of exposed dielectric surfaces.

(82) Semiconductor lithography is the formation of three-dimensional relief images or patterns on the semiconductor substrate for subsequent transfer of the pattern to the substrate. In semiconductor lithography, the patterns are formed by a light sensitive polymer called a photoresist. To build the complex structures that make up a transistor and the many wires that connect the millions of transistors of a circuit, lithography and etch pattern transfer steps are repeated multiple times. Each pattern being printed on the wafer is aligned to the previously formed patterns and slowly the conductors, insulators and selectively doped regions are built up to form the final device.

(83) The photoresist can be formed using conventional deposition techniques such chemical vapor deposition, plasma vapor deposition, sputtering, dip coating, spin-on coating, brushing, spraying and other like deposition techniques can be employed. Following formation of the photoresist, the photoresist is exposed to a desired pattern of radiation such as X-ray radiation, extreme ultraviolet (EUV) radiation, electron beam radiation or the like. Next, the exposed photoresist is developed utilizing a conventional resist development process.

(84) After the development step, the etching step can be performed to transfer the pattern from the patterned photoresist into the interlayer dielectric. The etching step used in forming the at least one opening can include a dry etching process (including, for example, reactive ion etching, ion beam etching, plasma etching or laser ablation), a wet chemical etching process or any combination thereof.

(85) For the sake of brevity, conventional techniques related to making and using aspects of the invention may or may not be described in detail herein. In particular, various aspects of computing systems and specific computer programs to implement the various technical features described herein are well known. Accordingly, in the interest of brevity, many conventional implementation details are only mentioned briefly herein or are omitted entirely without providing the well-known system and/or process details.

(86) In some embodiments, various functions or acts can take place at a given location and/or in connection with the operation of one or more apparatuses or systems. In some embodiments, a portion of a given function or act can be performed at a first device or location, and the remainder of the function or act can be performed at one or more additional devices or locations.

(87) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(88) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The present disclosure has been presented for purposes of illustration and description but is not intended to be exhaustive or limited to the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiments were chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

(89) The diagrams depicted herein are illustrative. There can be many variations to the diagram or the steps (or operations) described therein without departing from the spirit of the disclosure. For instance, the actions can be performed in a differing order or actions can be added, deleted or modified. Also, the term “coupled” describes having a signal path between two elements and does not imply a direct connection between the elements with no intervening elements/connections therebetween. All of these variations are considered a part of the present disclosure.

(90) The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

(91) Additionally, the term “exemplary” is used herein to mean “serving as an example, instance or illustration.” Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms “at least one” and “one or more” are understood to include any integer number greater than or equal to one, i.e., one, two, three, four, etc. The terms “a plurality” are understood to include any integer number

greater than or equal to two, i.e., two, three, four, five, etc. The term “connection” can include both an indirect “connection” and a direct “connection.”

(92) The terms “about,” “substantially,” “approximately,” and variations thereof, are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of $\pm 8\%$ or 5% , or 2% of a given value.

(93) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments described herein.

Claims

1. A semiconductor structure comprising: a transistor coupled to a memory element, the memory element being in series with a first bistable resistive element that is configured to switch between a first low resistance state and a first high resistance state; and a logic circuit coupled to the transistor via a series connection to a second bistable resistive element, the second bistable resistive element being configured to switch between a second low resistance state and a second high resistance state.
2. The semiconductor structure of claim 1, wherein the memory element is configured to form a circuit with the transistor in which the circuit electrically isolates the logic circuit.
3. The semiconductor structure of claim 1, wherein the logic circuit is configured to form a circuit with the transistor in which the circuit electrically isolates the memory element.
4. The semiconductor structure of claim 1, wherein the memory element is coupled to a memory array circuit in a back-end-of-line network on a front side, the front side being above the transistor.
5. The semiconductor structure of claim 1, wherein the logic circuit is part of a network for routing logic functions and delivering power.
6. The semiconductor structure of claim 1, wherein the memory element and the logic circuit are formed on opposite sides of the transistor.
7. The semiconductor structure of claim 1, wherein a circuit with the transistor and the memory element electrically isolates a terminal of the transistor that connects to the logic circuit.
8. The semiconductor structure of claim 1, wherein a circuit with the transistor and the logic circuit electrically isolates a terminal of the transistor that connects to the memory element.
9. The semiconductor structure of claim 1, wherein the first and second bistable resistive elements each comprise a unipolar, abrupt, reversible, and electrically triggered resistance switch that operates between two stable resistance states.
10. The semiconductor structure of claim 1, wherein at least one of the first and second bistable resistive elements comprises an insulator-to-metal transition material.
11. The semiconductor structure of claim 10, wherein the insulator-to-metal transition material comprises $\text{VO}_{0.2}$, $\text{NbO}_{0.2}$, $\text{Ca}_{0.2}\text{RuO}_{0.4}$, $\text{LaCoO}_{0.3}$, $\text{Ti}_{0.2}\text{O}_{0.3}$, $\text{Ti}_{0.3}\text{O}_{0.5}$, $\text{SmNiO}_{0.3}$, $\text{NdNiO}_{0.3}$, $\text{V}_{0.2}\text{O}_{0.3}$, $\text{V}_{0.4}\text{O}_{0.7}$, $\text{Fe}_{0.3}\text{O}_{0.4}$, an oxide of the form $\text{ABO}_{0.3}$ -perovskite, or any combinations thereof.
12. The semiconductor structure of claim 1, wherein at least one of the first and second bistable resistive elements comprises a threshold-switching selector formed by a combination of a thin insulator layer with a metal.
13. The semiconductor structure of claim 12, wherein the threshold-switching selector comprises Ag-doped $\text{HfO}_{0.2}$, $\text{Cu/HfO}_{0.2}$, $\text{Ag/TiO}_{0.2}$, Cu_xS , Ag/a-Si, $\text{AgTe/TiN/TiO}_{0.2}/\text{TiN}$,

W/Cu.sub.xS, or any combinations thereof.

14. The semiconductor structure of claim 1, wherein the memory element comprises non-volatile memory (NVM).

15. The semiconductor structure of claim 1, wherein the memory element comprises charge trap memory, magnetic random access memory (MRAM), ferroelectric-RAM (FRAM), phase-change memory (PCM), resistive-RAM (ReRAM), or any combinations thereof.

16. The semiconductor structure of claim 1, wherein the transistor is one of a planar field-effect transistor (FET), a finFET, gate-all-around (GAA) FET, a vertical-transport-FET (VTFET), and a stacked-FET.

17. The semiconductor structure of claim 1, wherein: a first mode, to selectively operate the memory element, comprises the first bistable resistive element in the first low resistance state and the second bistable resistive element in the second high resistance state; and a second mode, to selectively operate the logic circuit, comprises the first bistable resistive element in the first high resistance state and the second bistable resistive element in the second low resistance state, the first and second low resistance states permitting a flow of electrical current, the first and second high resistance states limiting a flow of electrical current.

18. A method comprising: providing a transistor coupled to a memory element, the memory element being in series with a first bistable resistive element that is configured to switch between a first low resistance state and a first high resistance state; and providing a logic circuit coupled to the transistor via a series connection to a second bistable resistive element, the second bistable resistive element being configured to switch between a second low resistance state and a second high resistance state.

19. The method of claim 18, wherein: the memory element is configured to form a circuit with the transistor in which the circuit electrically isolates the logic circuit; and the logic circuit is configured to form another circuit with the transistor in which the another circuit electrically isolates the memory element.

20. The method of claim 18, wherein: a first mode, to selectively operate the memory element, comprises the first bistable resistive element in the first low resistance state and the second bistable resistive element in the second high resistance state; and a second mode, to selectively operate the logic circuit, comprises the first bistable resistive element in the first high resistance state and the second bistable resistive element in the second low resistance state, the first and second low resistance states permitting a flow of electrical current, the first and second high resistance states limiting a flow of electrical current.
